RAG Course FAQ - VPRO Skills

Frequently asked questions about RAG, Adaptive RAG, Hybrid RAG, ChromaDB, embeddings and OpenAI.

Q1. What does RAG stand for?

A: RAG stands for Retrieval-Augmented Generation. A RAG system retrieves relevant passages from a knowledge base and passes them to a large language model so that the answer is grounded in those passages.

Q2. Why not just ask the LLM directly?

A: An LLM only knows what was in its training data. It does not know your private course notes or recent changes, and it may hallucinate. RAG supplies the needed facts at question time and allows citations.

Q3. What is Adaptive RAG?

A: Adaptive RAG classifies each query and chooses a strategy: answer directly (DIRECT), retrieve from documents (RETRIEVE), or ask a follow-up question (CLARIFY). It avoids unnecessary retrieval for simple messages.

Q4. When does the assistant ask a clarifying question?

A: When a question is incomplete or ambiguous and the conversation history does not resolve it, for example 'Explain it' as the very first message. Clarifying avoids answering the wrong question.

Q5. What is Hybrid RAG?

A: Hybrid RAG combines sparse keyword retrieval such as BM25 with dense embedding retrieval and fuses the results, often using Reciprocal Rank Fusion. It handles both exact terms and semantic matches.

Q6. What is ChromaDB?

A: ChromaDB is an open-source vector database. It stores embeddings, chunk text, and metadata, supports similarity search, and can persist data to a local directory so the index survives restarts.

Q7. Where does this project store its ChromaDB data?

A: In a persistent local directory configured by CHROMA_PERSIST_DIRECTORY, by default backend/storage/chroma. The collection name is configured by CHROMA_COLLECTION_NAME.

Q8. What are embeddings?

A: Embeddings are numeric vectors that capture the meaning of text. Similar texts have similar vectors, so comparing vectors with cosine similarity finds passages related to a question.

Q9. Which OpenAI models does the course project use?

A: By default it uses gpt-4.1-mini for routing and answer generation and text-embedding-3-small for embeddings. Both are configurable with OPENAI_CHAT_MODEL and OPENAI_EMBEDDING_MODEL.

Q10. What is the relevance threshold?

A: It is the minimum cosine similarity a retrieved chunk must have to be used as context. Chunks below the threshold are ignored; if none pass, the assistant says the documents do not contain the answer.

Q11. Does using the OpenAI API cost money?

A: Yes. Embedding documents and generating answers are billed by OpenAI per token. Indexing the small

sample documents costs very little, and unchanged files are not re-embedded.

Q12. What is chunk overlap?

A: Chunk overlap repeats a small number of characters between consecutive chunks so that information near a chunk boundary is not lost. This project uses 120 characters of overlap with 800-character chunks.